

Superposition of Two Plane Polarized Waves

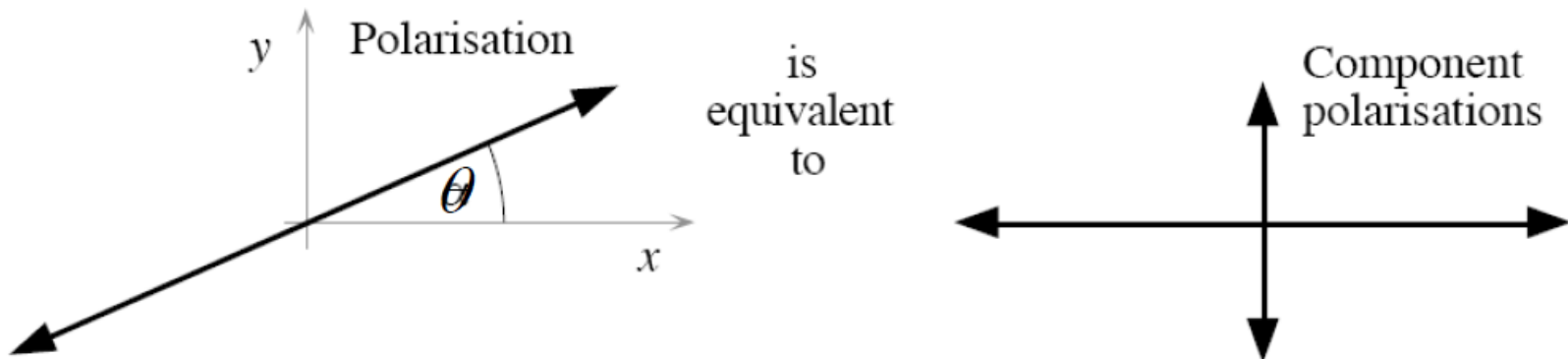
Let a beam of plane polarized light is incident normally on a calcite plate cut with its optic axis parallel to its face (x-axis).

It will split into two orthogonal waves known as O-ray and E-ray , having perpendicular electric fields . The electric field associated with the two orthogonal waves (O-ray and E-ray in Anisotropic crystal) can be written as

$$E_x(z,t) = E_{0x} \sin(kz - \omega t)$$

$$E_y(z,t) = E_{0y} \sin(kz - \omega t + \phi)$$

ϕ is the phase difference introduced between the two orthogonal component of the incident ray.



Superposition of Two Plane Polarized Waves

$$E_x(z, t) = E_{0x} \sin(kz - \omega t) \dots (1)$$

$$E_y(z, t) = E_{0y} \sin(kz - \omega t + \phi) \dots (2)$$

From equation (1) and (2)

$$\frac{E_x(z, t)}{E_{0x}} = \sin(kz - \omega t) \dots (3) \text{ and}$$

$$\frac{E_y(z, t)}{E_{0y}} = \sin(kz - \omega t + \phi) \dots (4)$$

$$\frac{E_y(z, t)}{E_{0y}} = \sin(kz - \omega t) \cos \phi + \cos(kz - \omega t) \sin \phi$$

$$\frac{E_y(z, t)}{E_{0y}} = \frac{E_x(z, t)}{E_{0x}} \cos \phi + \sin \phi \sqrt{1 - \frac{E_x^2(z, t)}{E_{0x}^2}}$$

$$\Rightarrow \frac{E_y}{E_{0y}} - \frac{E_x}{E_{0x}} \cos \phi = \sin \phi \sqrt{1 - \frac{E_x^2}{E_{0x}^2}}$$

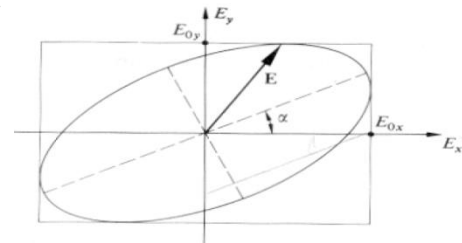
Squaring both side

$$\Rightarrow \frac{E_y^2}{E_{0y}^2} + \frac{E_x^2}{E_{0x}^2} - 2 \frac{E_x}{E_{0x}} \frac{E_y}{E_{0y}} \cos \phi = \sin^2 \phi$$

This is the equation of the ellipse tilted at an angle α which is made by the major or minor axis of the ellipse with the x axis.

This angle α is given by

$$\tan 2\alpha = \frac{2E_{0x}E_{0y} \cos \phi}{E_{0x}^2 - E_{0y}^2}$$

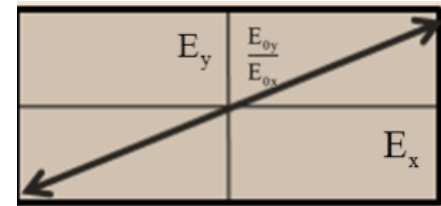


Nature of emerging light will depend on two factors:

- phase difference between two components
- initial amplitude of two components

1) When $\phi = 0, 2\pi, 4\pi, \dots$ or $2n\pi$

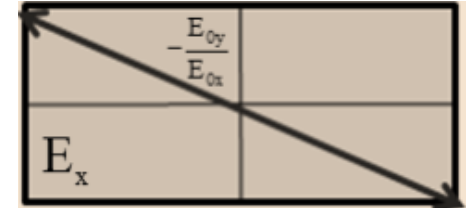
$$\Rightarrow E_y = \frac{E_{oy}}{E_{ox}} E_x$$



Thus emergent ray is a linearly polarized light with vibrations in the same plane as of incident light.

2) When $\phi = \pi, 3\pi, 5\pi, \dots$ or $(2n+1)\pi$

$$\Rightarrow E_y = -\frac{E_{oy}}{E_{ox}} E_x$$



This is again a straight line showing that emergent light will be a linearly polarized light.

3) When $\phi = \pm(2n+1)\pi/2$

This is a equation of simple ellipse. Thus emergent light is elliptically polarized.

$$\frac{E_x^2}{E_{0x}^2} + \frac{E_y^2}{E_{0y}^2} = 1$$

4) When $\phi = \pm(2n+1)\pi/2$ and $E_{oy} = E_{ox} = E_0$

$$E_x^2 + E_y^2 = E_0^2$$

This is a equation of circle. Thus emergent light would be circularly polarized.

Note: +ve values of for RCP or REP and -ve for LCP or LEP

Problems

Q.1: Explain the state of polarizations in the following equations:

(i) $E_x = E_o \cos(\omega t + kz)$; $E_y = E_o / \sqrt{2} \cos(\omega t + kz + \pi)$

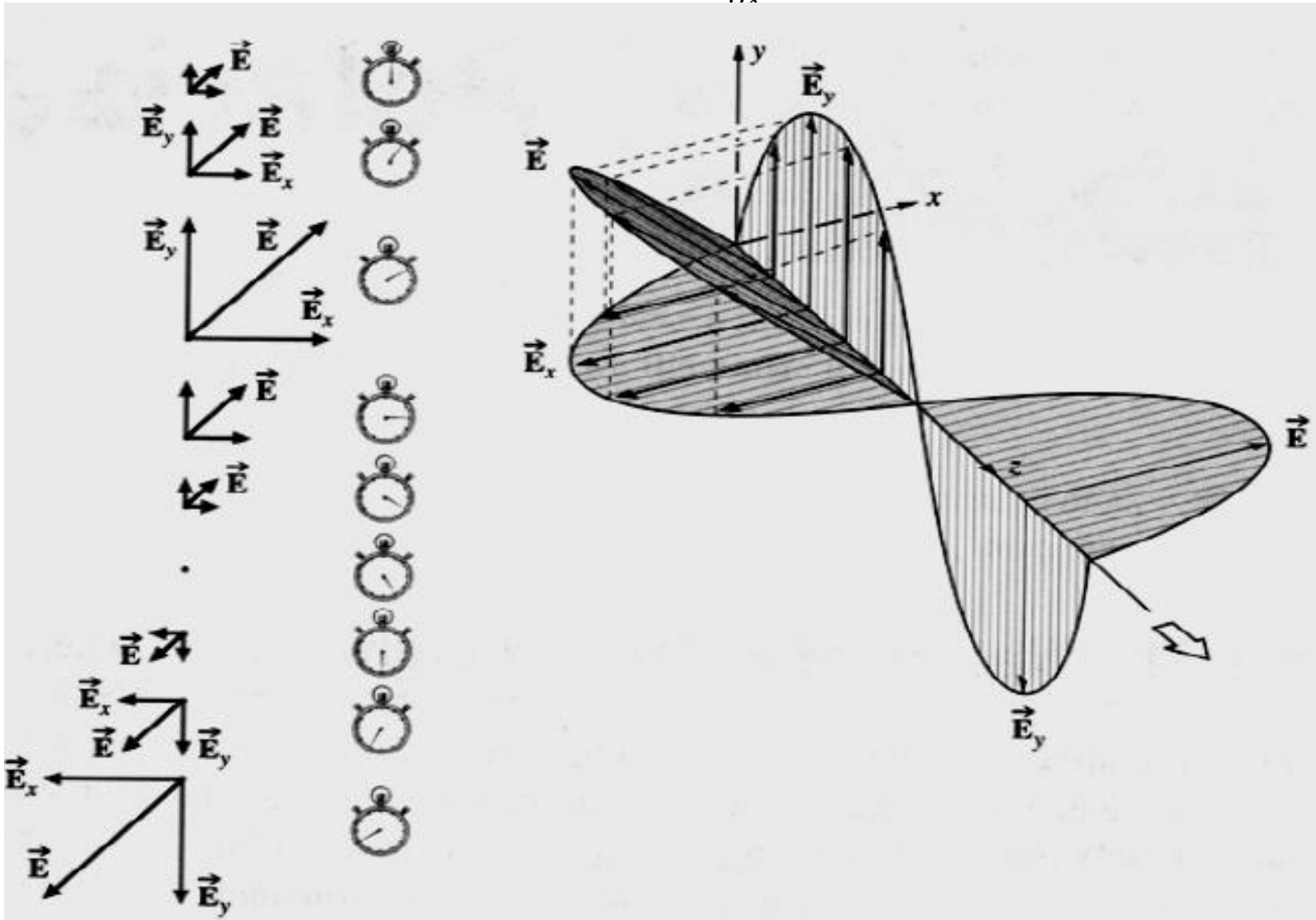
(ii) $E_x = E_o \sin(\omega t + kz)$; $E_y = E_o \cos(\omega t + kz + \pi)$

(iii) $E_x = E_o \sin(kz - \omega t + \pi/3)$; $E_y = E_o \sin(kz - \omega t - \pi/6)$

(iv) $E_x = E_o \sin(kz - \omega t + \pi/4)$; $E_y = E_o / \sqrt{2} \sin(kz - \omega t)$

Graphical Representations

If $\theta = 0, 2\pi, 4\pi, 6\pi, \dots$ or $2n\pi \Rightarrow E_y = \frac{E_{oy}}{E_{ox}} E_x$



Circularly Polarized Light

$$\text{If } \theta = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots = \pm(2n+1)\frac{\pi}{2} \text{ and } E_{0x} = E_{0y} = E_0$$

$$E_x(z, t) = E_0 \cos(kz - \omega t)$$

at $z=0$

$$E_x(t) = E_0 \cos(\omega t)$$

$$E_y(z, t) = E_0 \cos(kz - \omega t + \phi)$$

$$E_y(t) = E_0 \cos(\phi - \omega t)$$

For $\phi = +\pi/2$

$$E_x(t) = E_0 \cos \omega t$$

$$E_y(t) = E_0 \sin \omega t$$

Now squaring and adding E_x and E_y

$$E_x^2 + E_y^2 = E_0^2$$

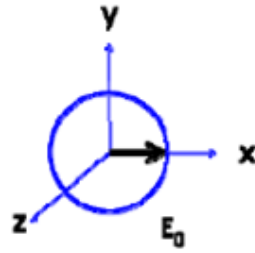
Circularly Polarized Light

$$E_x(t) = E_0 \cos \omega t \quad E_y(t) = E_0 \sin \omega t$$

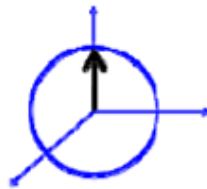
For $z = 0$ and $\phi = -\pi/2$

$$E_x(t) = E_0 \cos \omega t \quad E_y(t) = -E_0 \sin \omega t$$

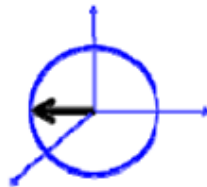
At $t = 0, E_x = E_0; E_y = 0$



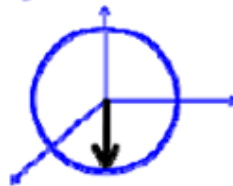
At $t = \frac{T}{4}, E_x = 0; E_y = E_0$



At $t = \frac{T}{2}, E_x = -E_0; E_y = 0$

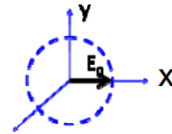


At $t = \frac{3T}{4}, E_x = 0; E_y = -E_0$

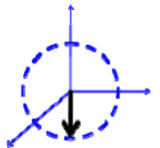


Left Circularly Polarized Light (LCP)

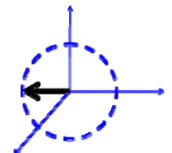
At $t = 0, E_x = E_0; E_y = 0$



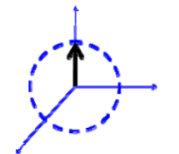
At $t = \frac{T}{4}, E_x = 0; E_y = -E_0$



At $t = \frac{T}{2}, E_x = -E_0; E_y = 0$

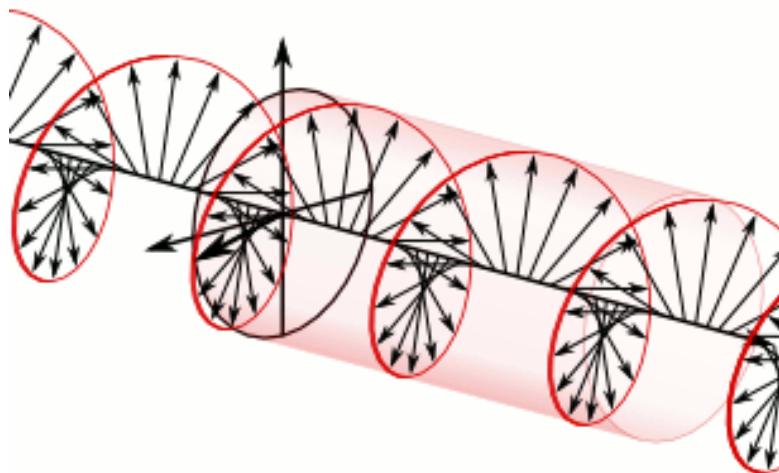


At $t = \frac{3T}{4}, E_x = 0; E_y = E_0$

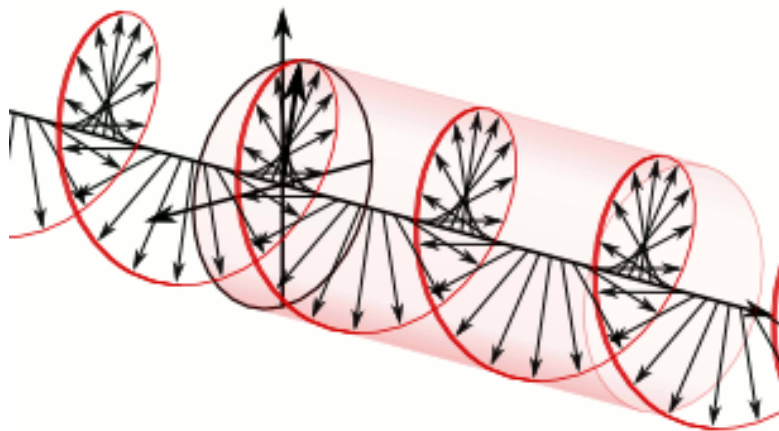


Right Circularly Polarized Light (RCP)

Circularly Polarized Light

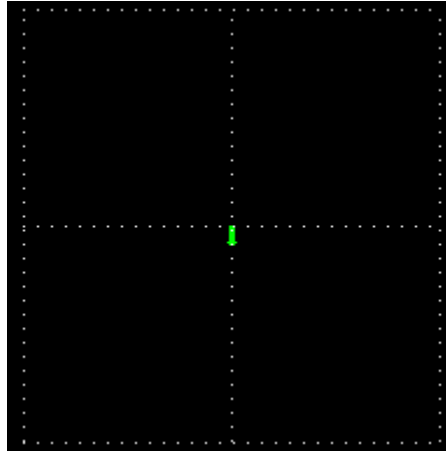
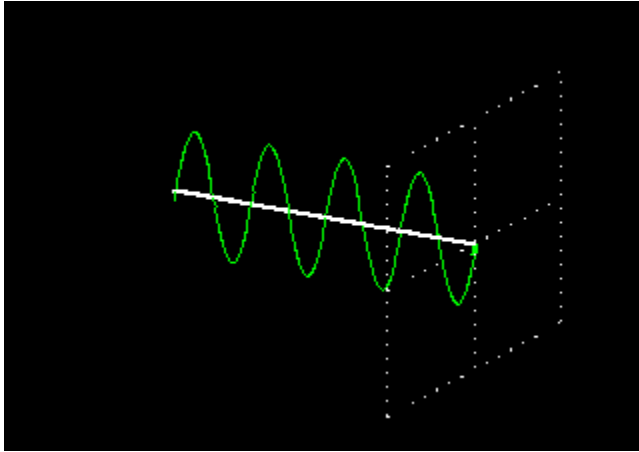


A right-handed/clockwise circularly polarized wave as defined from the point of view of the source. It would be considered left-handed/anti-clockwise circularly polarized if defined from the point of view of the receiver.

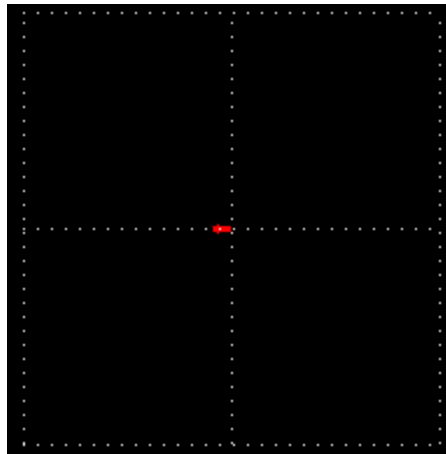
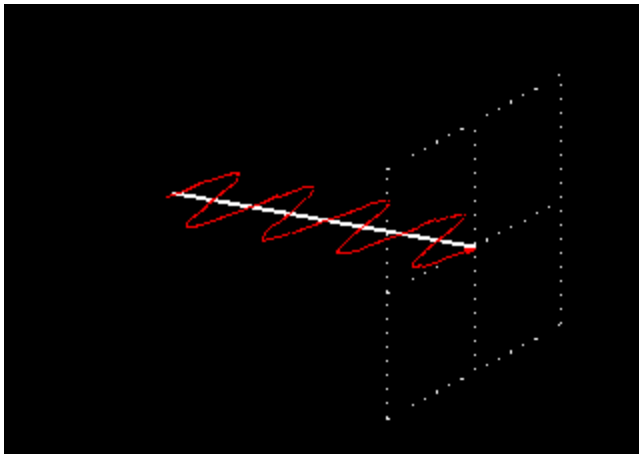


A left-handed/anti-clockwise circularly polarized wave as defined from the point of view of the source. It would be considered right-handed/clockwise circularly polarized if defined from the point of view of the receiver.

Plane-polarized waves



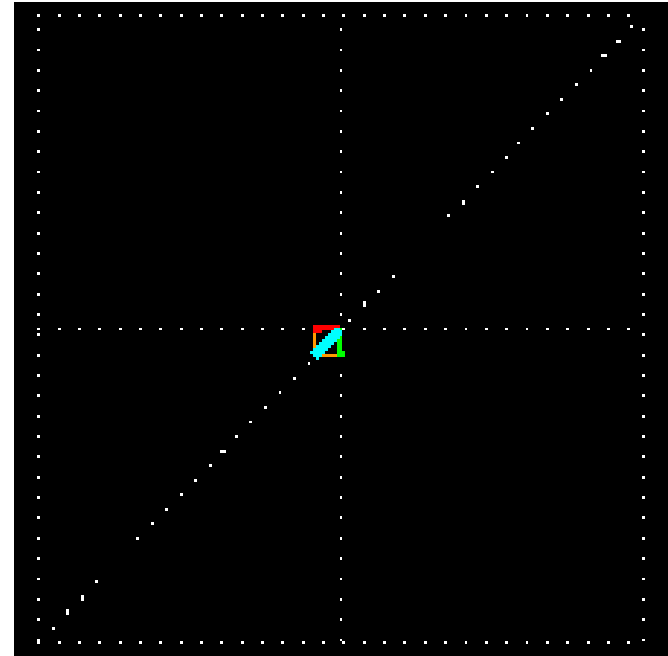
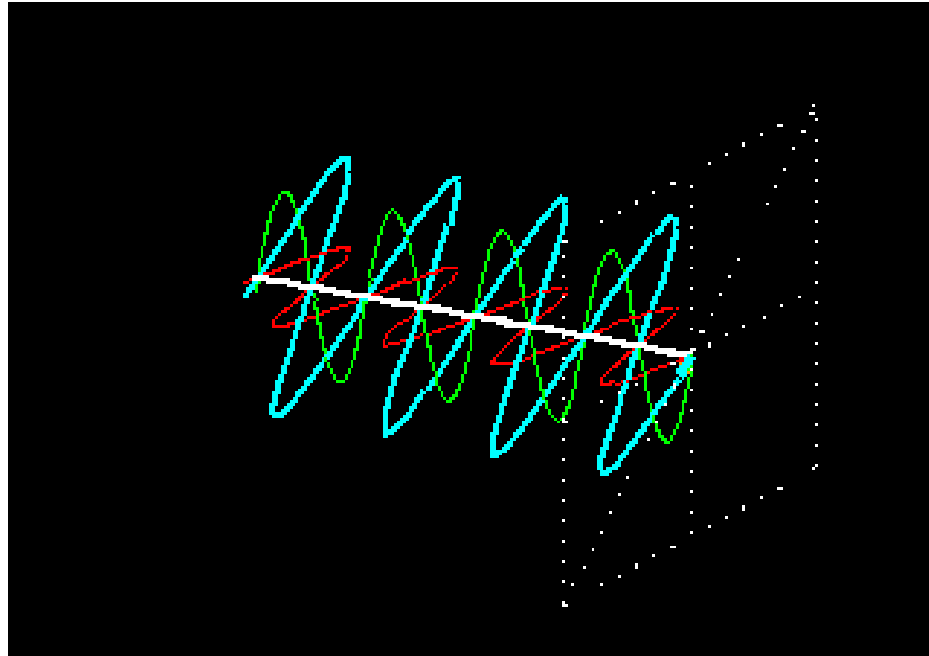
If the vector of the electric field (measured at a fixed point of space) oscillates along a straight line then the waves are called **plane-polarized** or **linearly polarized** waves.



3D view

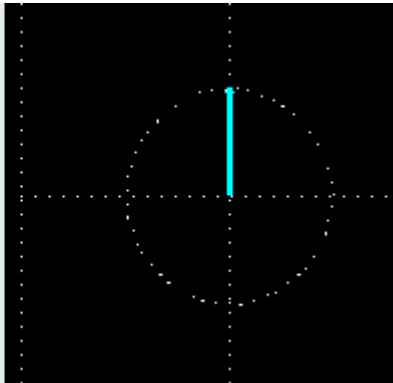
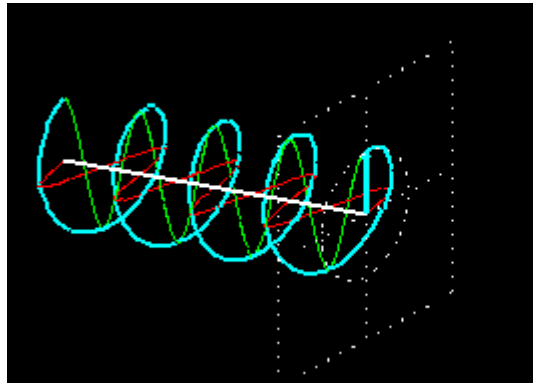
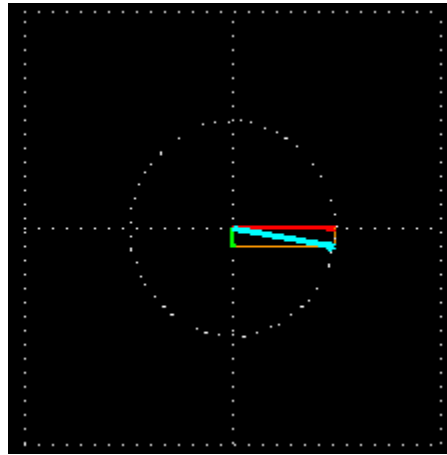
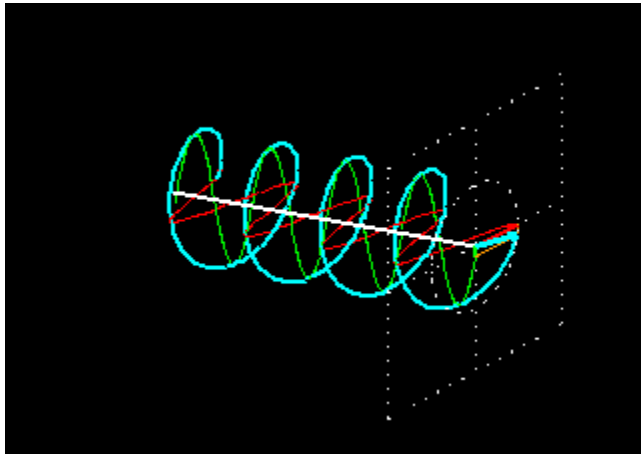
Front view

Superposition of plane-polarized waves



The following animation presents the superposition of two waves that have the same amplitude and wavelength, are polarized in two perpendicular planes and oscillate in *the same phase*. Oscillating in the same phase means that the two waves reach their peaks and cross the zero line in the same moments. The superposing components are visualized with red and green color, respectively. The wave resulting from the superposition is shown in light blue.

Superposition of plane-polarized waves



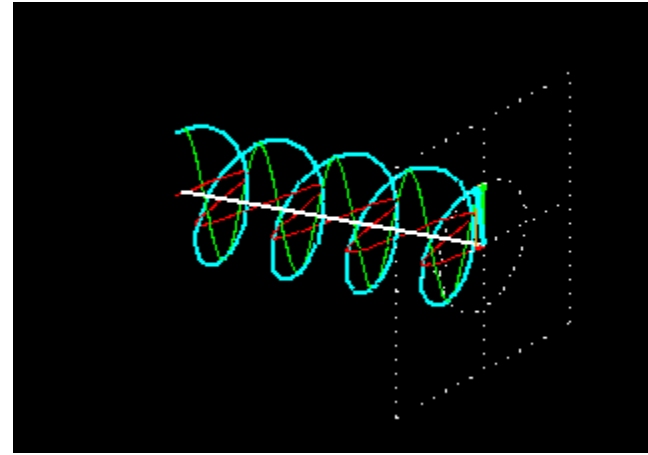
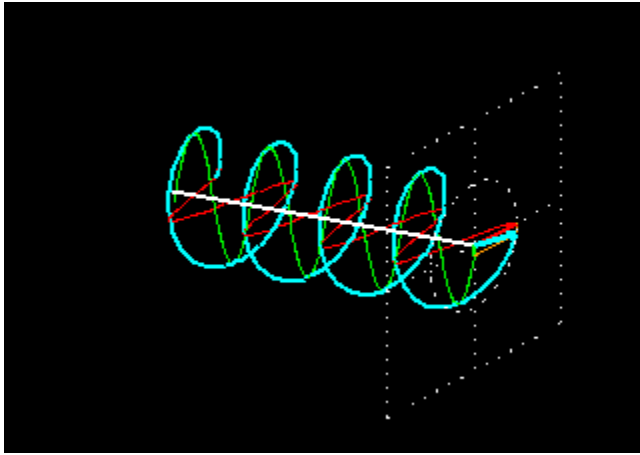
When two waves plane-polarized in two perpendicular planes meet **out of phase** then the wave resulting from the superposition of the two waves will no longer be plane-polarized. The following animation presents the superposition of two waves that have the same amplitude and wavelength and are polarized in two perpendicular planes but there is a phase difference of 90° between them. A phase difference of 90° means that when one wave is at its peak then the other one is just crossing the zero line.

At any fixed point in space that is in the line of the propagation of this wave, the electric field vector *rotates in a circle while its length remains constant*. Such waves are called **circularly polarized waves**.

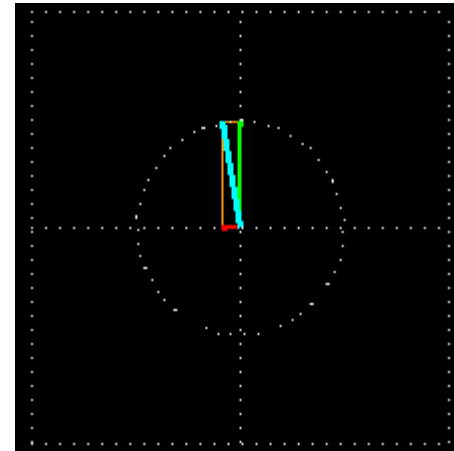
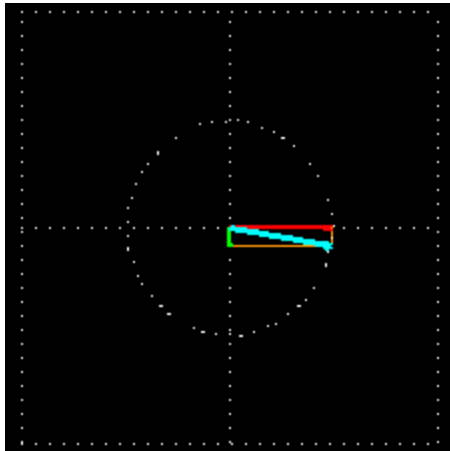
As the 3D picture shows, a circularly polarized wave can be visualized with a spiral line; the wave propagates as a function describing a spiral instead of one describing a sine curve.

In this picture, **the electric field vector rotates clockwise** when looked at from the direction of propagation.

Circularly Polarized Light



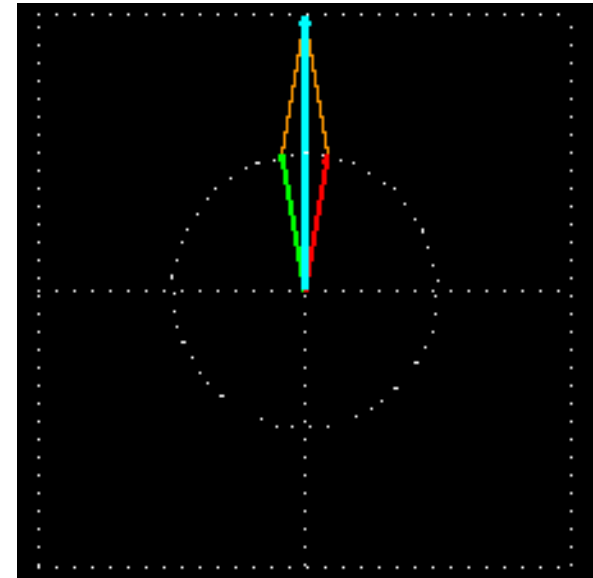
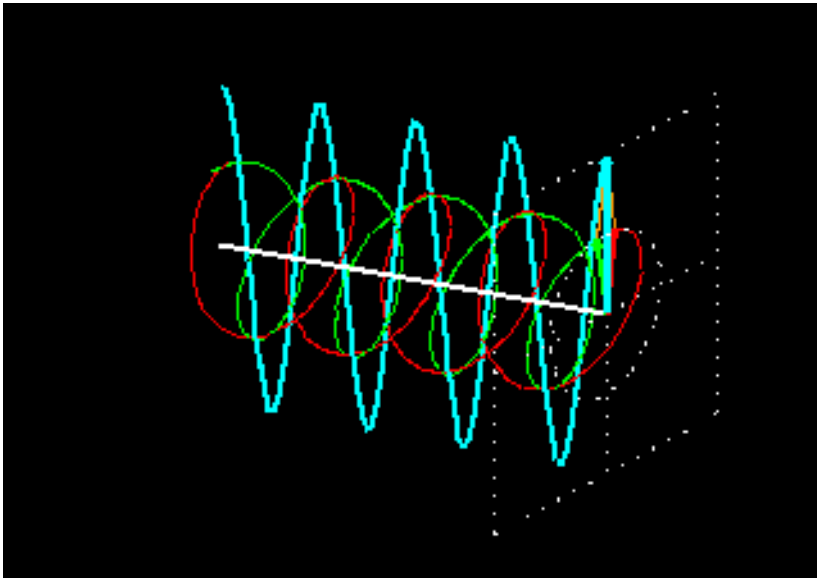
When the intersecting planes are looked at from the front then the following images are seen:



Superposition of Circularly Polarized Waves

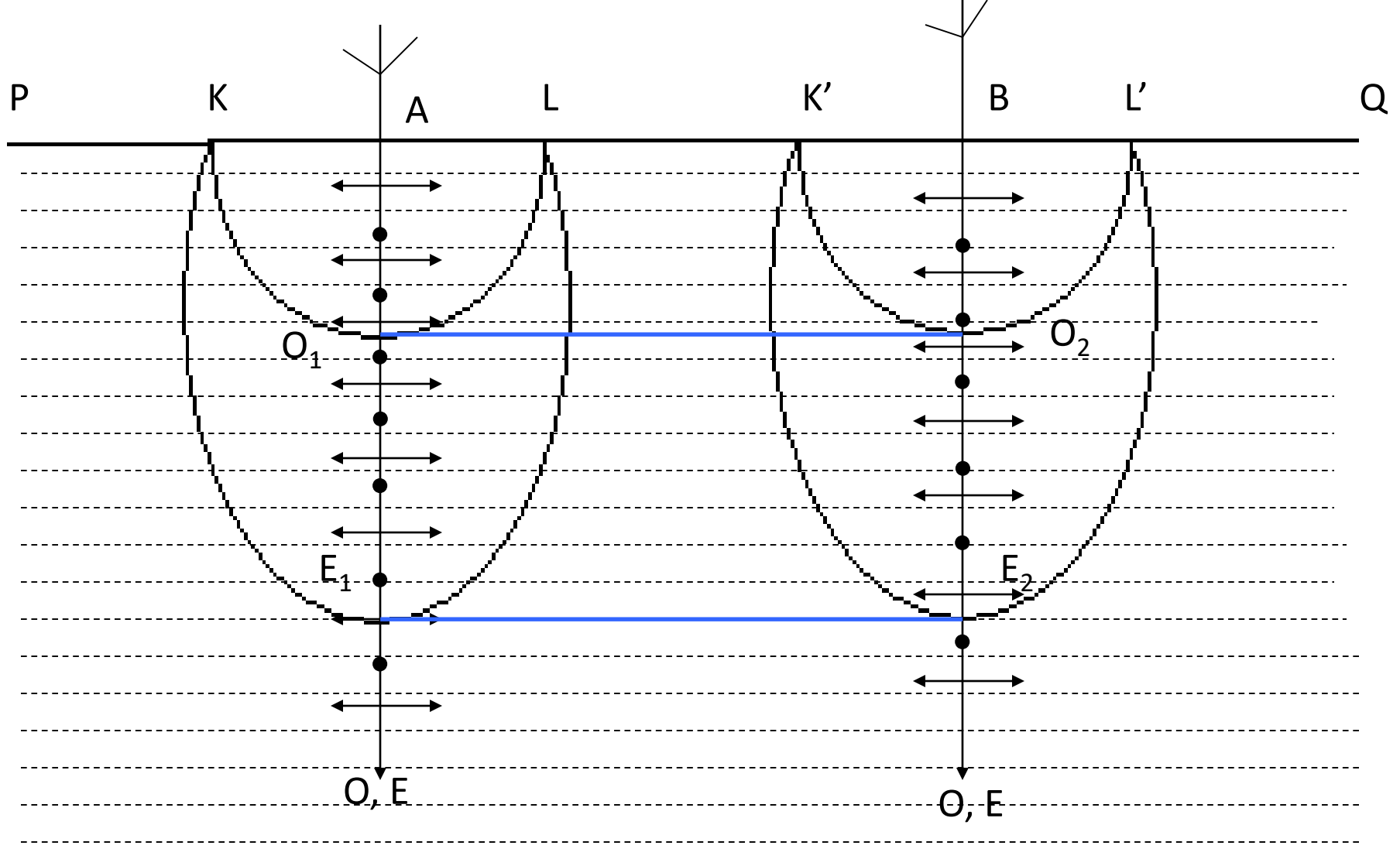
What will happen when a left circularly polarized wave and a right circularly polarized wave are added?

Let's assume that the two light beams have equal amplitudes and wavelengths.



The result of superposing two circularly polarized waves is a *plane-polarized wave*.

Any linearly polarized light wave can be obtained as a superposition of a left circularly polarized and a right circularly polarized light wave, whose amplitudes are identical.



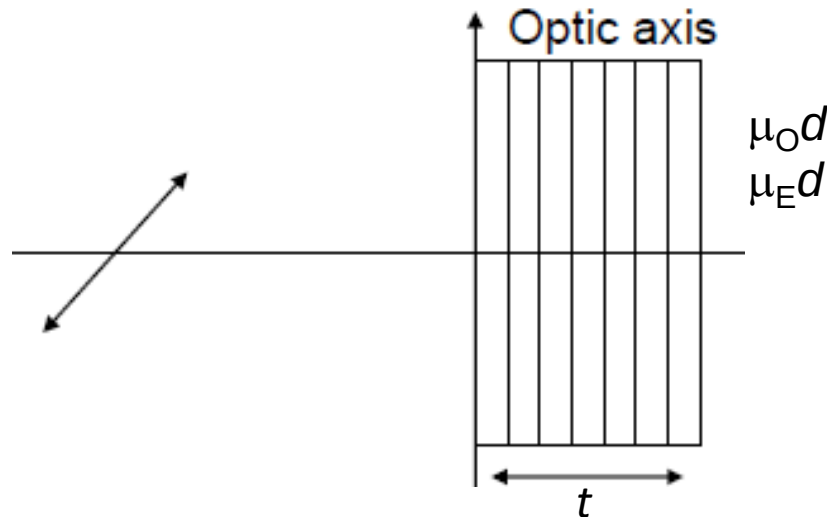
Optic axis parallel to refracting surface, maximum bi-refringence with same path. Hence it can be used to make wave *retarders*.

Retarders

This is an optical device, which introduces a phase difference between extra-ordinary and ordinary rays.

These are in the form of plates of doubly refracting crystal cut in such a way that optic axis is parallel to the refracting surfaces.

Path difference between E-wave & O-wave is $(\mu_O - \mu_E)d$.



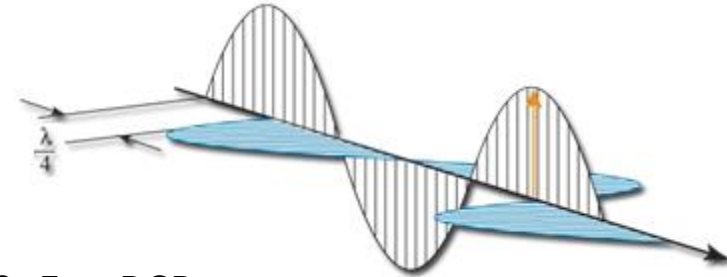
Quarter-Wave Plate

If doubly refracting crystal is having thickness d such that path difference between E-ray and O-ray is $\lambda/4$ i.e.

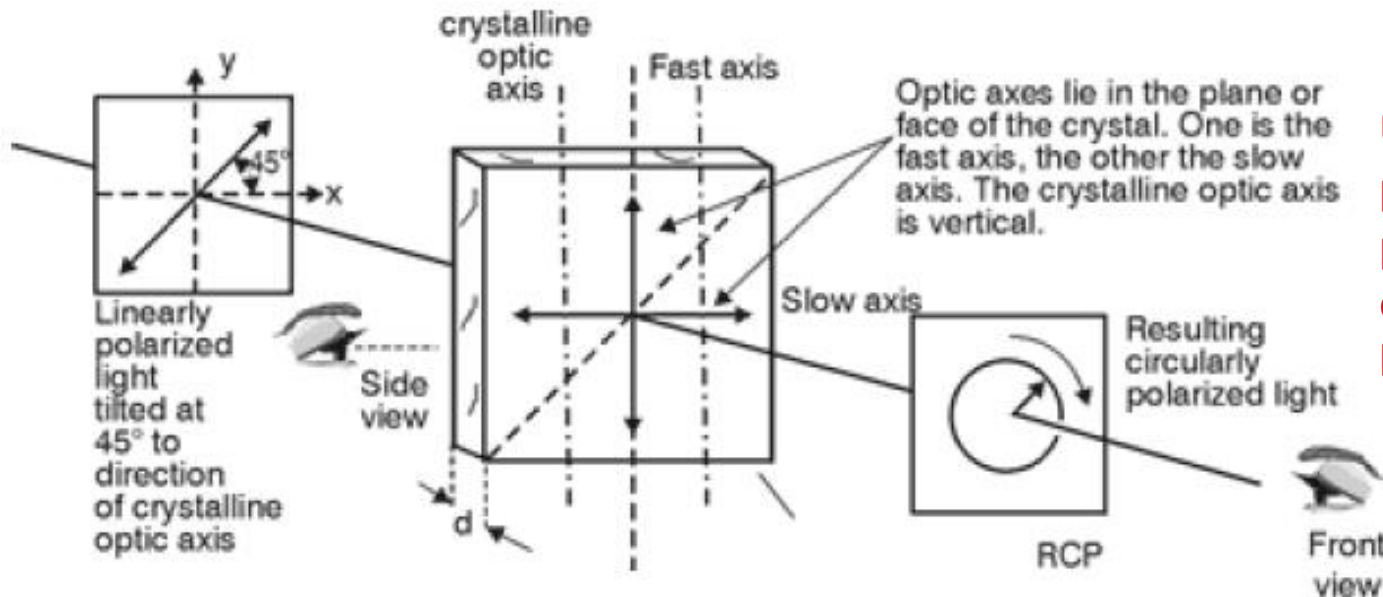
$$(\mu_E \sim \mu_O)d = (2n+1)\frac{\lambda}{4} \quad n = 0, 1, 2, 3, \dots$$

The crystal is known as Quarter-Wave Plate (QWP).

$$\Rightarrow d = \frac{\lambda}{4(\mu_E \sim \mu_O)} \quad \phi = (2n+1)\frac{\pi}{2}$$



For $n = 0, 2, 4, \dots$ emergent light will be LCP and for $n = 1, 3, 5, \dots$ RCP



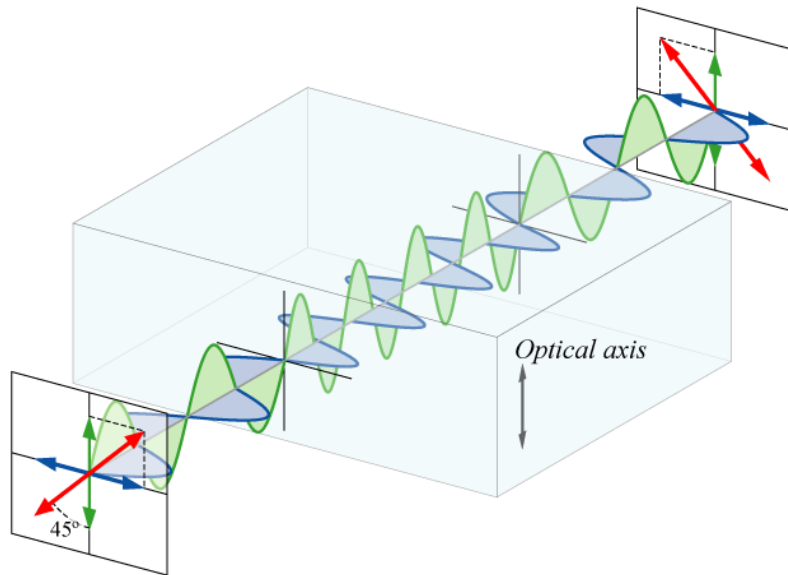
Use: QWP Convert plane polarized (PP) to circular polarized (CP) or elliptically polarized (EP) light and vice versa.

Half-Wave Plate

If doubly refracting crystal is having thickness t such that path difference between E-ray and O-ray is $\lambda/2$ i.e.

$$(\mu_E \sim \mu_O)d = (2n+1)\frac{\lambda}{2} \quad n=0,1,2,3\dots$$
$$\Rightarrow d = \frac{\lambda}{2(\mu_E \sim \mu_O)}$$
$$\phi = (2n+1)\pi$$

Use: HWP Convert Right circular polarized (RCP) or right elliptically polarized (REP) light to LCP or LEP and vice versa.

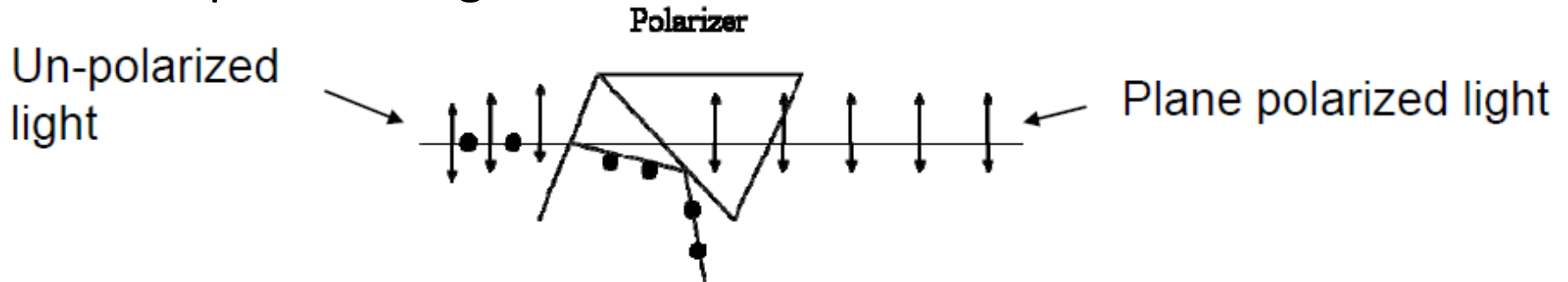


Linearly polarized light entering a half-wave plate can be resolved into two waves, parallel (shown as green) and perpendicular (blue) to the optical axis of the wave plate. In the plate, the parallel wave propagates slightly slower than the perpendicular one. At the far side of the plate, the parallel wave is exactly half of a wavelength delayed relative to the perpendicular wave, and the resulting combination (red) is orthogonally polarized compared to its entrance state.

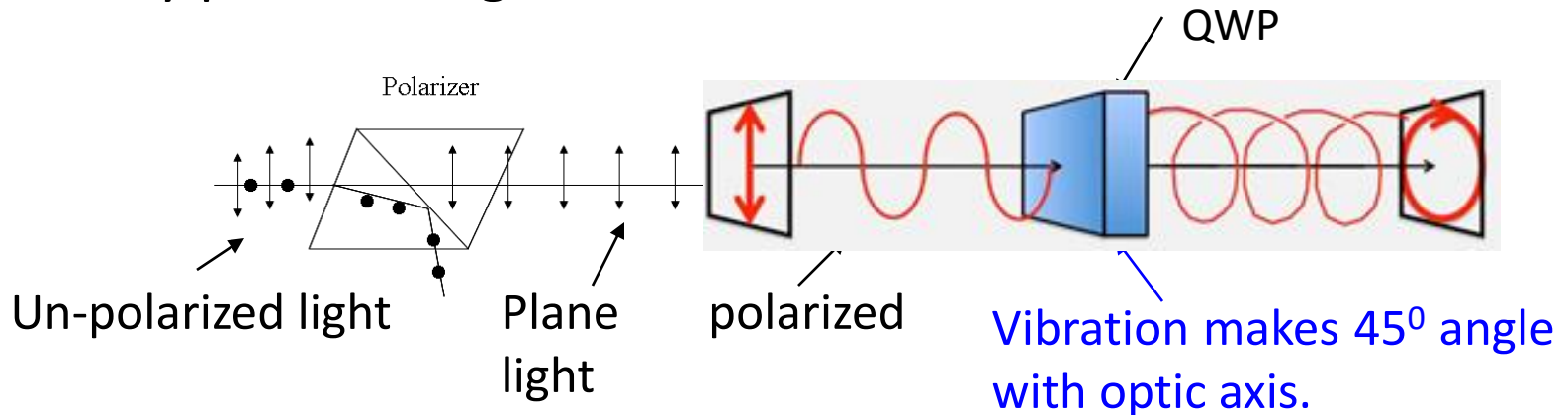
Similarly λ (Full wave Plate) , $\lambda/6$ plate or $\lambda/8$ plate etc

Production of Polarized Light

1. Plane polarized light:



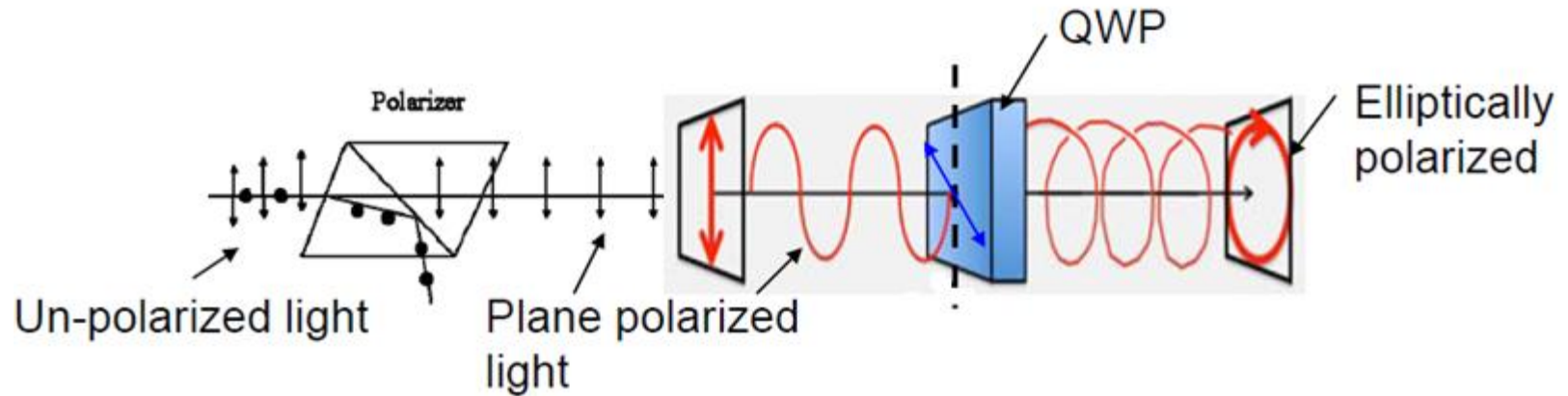
2. Circularly polarized light:



If quarter plate is at 45° , light passing through it will be circularly polarized.
No variation of intensity with the rotation of analyzer.

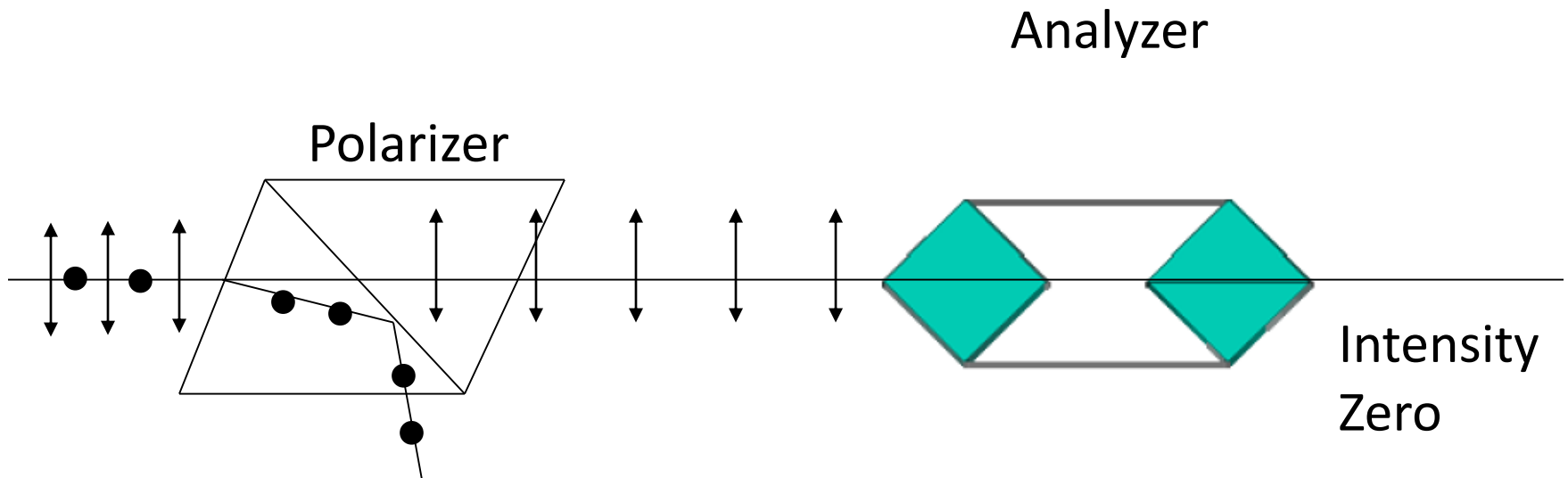
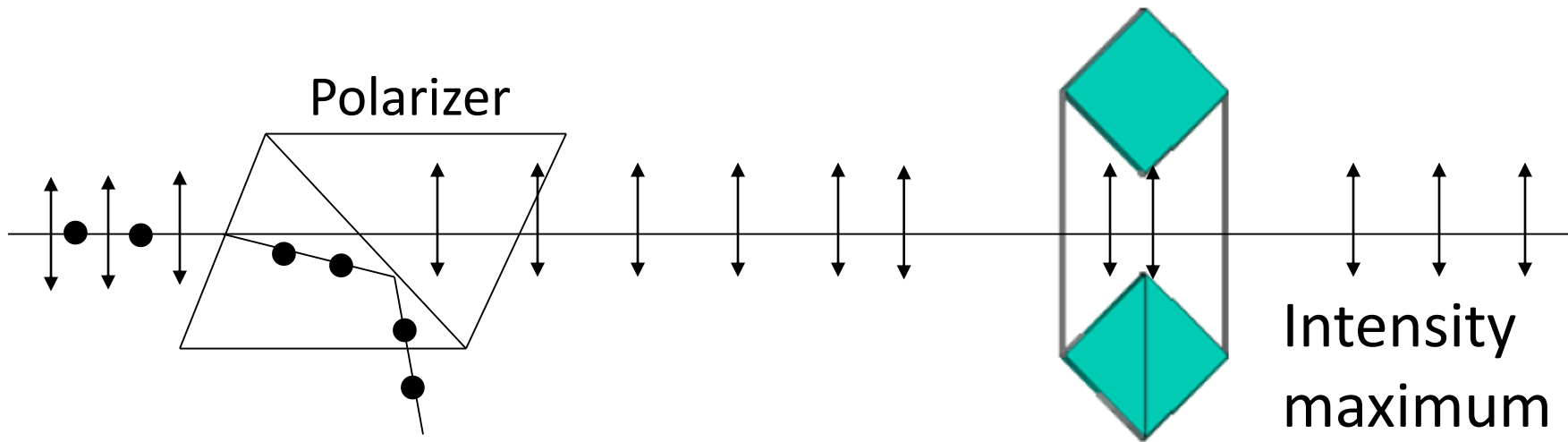
Production of Polarized Light

3. Elliptically polarized light:



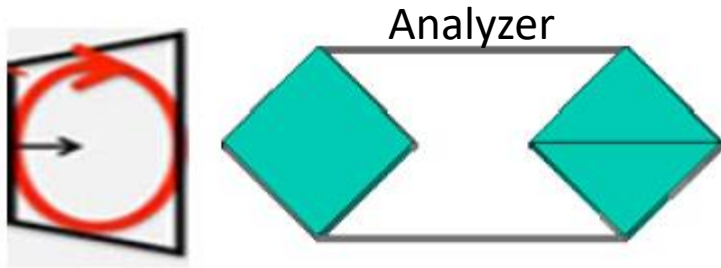
Analysis of Polarized Light

1. Plane polarized light:



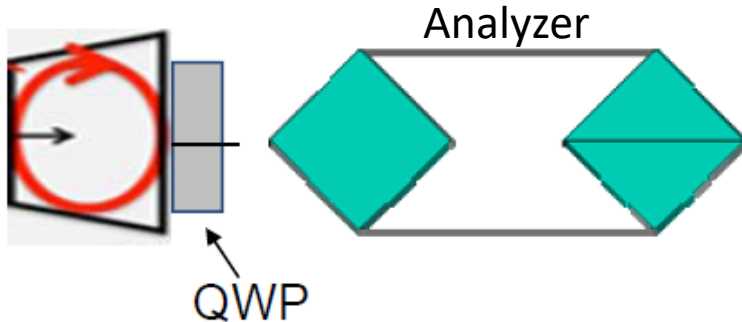
Analysis of Polarized Light

2. Circularly polarized light:



No variation
in intensity

It may be a unpolarized or
may be a circularly polarized light

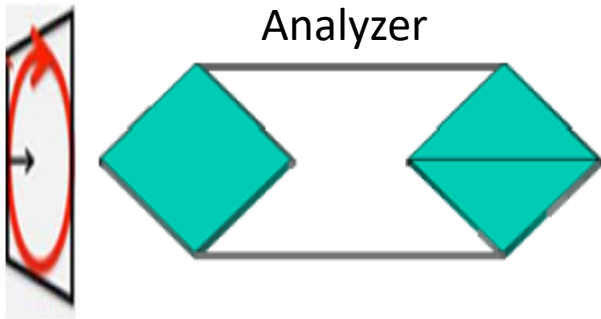


If variation in intensity is like plane polarized
light original light is circularly polarized.

Otherwise, original light is un-polarized.

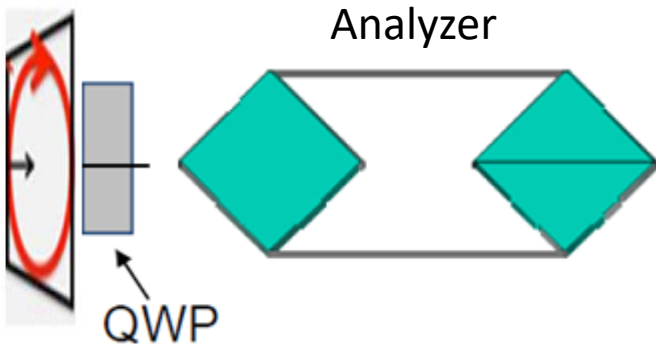
Analysis of Polarized Light

3. Elliptically polarized light:



Variation of intensity from a maximum to minimum

It may be a partially polarized or may be an elliptically polarized light



If variation in intensity is like plane polarized light original light is elliptically polarized.

Otherwise, original light is un-polarized.

Scheme of analysis of a given beam of light

Given beam of light

Incident on a rotating Nicol prism

Variation in intensity with minimum non zero

Conclusion: **Given light is either elliptically polarized or partially polarized**

Variation in intensity with minimum zero

Conclusion: **Given light is plane polarized**

No Variation in intensity

Conclusion: **Given light is either circularly polarized or unpolarized.**

Incident on a QWP with optic axis || to the pass axis of the analyzing Nicol at the position of maximum intensity and then examined by rotating Nicol prism

Variation in intensity with minimum zero

Conclusion: **elliptically polarized**

Variation in intensity with minimum non zero

Conclusion: **partially polarized**

Incident on a QWP in any position and then examined by rotating Nicol prism

Variation in intensity with minimum zero
Conclusion: **circularly polarized**

No Variation in intensity
Conclusion: **unpolarized.**